

Homework 1. Population of a metropolis

Let us imagine a hypothetical metropolis with two suburbs (U1, U2) and a city centre (C). Initially, the suburbs have populations of 100,000 and 50,000, and the city's total population ($U1 + U2 + C$) is 800,000. Migration rates between the various regions of the city on a year-to-year basis unfold as follows:

$$C \rightarrow U1: 0.2$$

$$U1 \rightarrow C: 0.1$$

$$C \rightarrow U2: 0.1$$

$$U2 \rightarrow C: 0.05$$

$$U1 \rightarrow U2: 0.3$$

$$U2 \rightarrow U1: 0.05$$

All other people move in the same part of the city or does not move anywhere.

Create a script to calculate population of the metropolis regions as a time function! Analyse the effect of $U1 \rightarrow U2$ transport rate on the steady state population distribution!

Extra problem (not part of HW1): Use a random variable-based migration to simulate the year-to-year migration. (i.e. A random number (x) is generated between 0 and 1. If population of city region is 120000 then $x \cdot 120000$ citizen will move to U1 region.) Compare the population diagram with the above Markov model's population!